

1.

Prompt the user to repeatedly to do the program(input(Yes or No)). If they respond Yes, go into the loop and prompt them for last name, month and sales. Write a function to compute next month's forecast. Pass to the function month and sales. Determine the forecast percent (see below) and compute next month's sales to be sales x (1+forecast percent). Return next month's sales and display the value.

Month/Forecast %

Jan, Feb, Mar

0.10

Apr, May, Jun

0.15

Jul, Aug, Sep

0.20

Oct, Nov, Dec

0.25

Input	Process	Output
Y/N	Pass function month and sales	Value for next month sales
Iname	Forecast % based on month	Name
Month	Next Mon sales = sales * (1+forecast %)	
# of Sales	Return Next Mon sales	

2.

Prompt the user to repeatedly to do the program(input (Yes or No)). If they response Yes go into the loop and prompt the user for length, width and height of a room. Write a function to

compute the square footage of the room. The function should receive the length, width and height of the room and return square footage ($2 \times \text{length} \times \text{width}$ (floor and ceiling) + $2 \times \text{length} \times \text{height}$ (2 of the walls) + $2 \times \text{width} \times \text{height}$ (the other 2 walls)). A gallon of paint covers 50 square feet. Compute the number of gallons needed to paint the room (square footage of the room / 50). Display the number of gallons needed.

Input	Process	Output
Y/N	Compute sq ft: $\text{Sq ft} = 2\text{LW} + 2\text{LH} + 2\text{WH}$	# of gallons needed
Length, Width, Height	Compute Gal needed: $\text{SQFT} / 50$	
	Return Gal needed	

3. Prompt the user to repeatedly to do the program(input (Yes or No)). If they response Yes go into the loop and prompt the user for make, model, electric vehicle code (Y or N) and MSRP(sticker price) of an automobile. Write a function to compute the out the door price. Pass to the function the MSRP, make, model and electric vehicle code. Determine the percent off the MSRP then compute the new MSRP and finally add 7% sales tax to the total. Return and display the total. Also sum all MSRP's and sum of all sales price of the cars (MSRP discount + tax).

To determine percent off MSRP/% off MSRp

Honda Accord

0.10

Toyota Rav4

0.15

All electric vehicles

0.30

All other vehicles

0.05

Input	Process	Output
Y/N	Pass msrp, make + model and EV code to function	Display final price
Make,Model,	Final Price = msrp * Discount % * 7% sales tax	Sum of all prices? (MSRP)
Electric Vehicle? Y/N	Sum all MSRP's	Make model?
MSRP	Sum all sales price of the cars (MSRP discount+tax)	

4.Prompt the user to repeatedly to do the program(input (Yes or No)). If they response Yes go into the loop and prompt the user for last name and miles from downtown Chicago. Write a function to compute the train ticket price. Pass to the function the mile from down town Chicago and determine the ticket price. Return the ticket price. Sum price of all tickets.

Miles from Down Town Chicago /Ticket Price

30 or more

\$12

20 to 29

\$10

10 to 19

\$8

All others (under 10 miles??)

\$5

Input	Process	Output
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Y/N	Pass miles to funcion	Iname?
Iname	30m+ = \$12 20-29m = \$10 10-19 = \$8 10m> = \$5	Ticket price total
Miles of the trip	Return ticket price	Ticket price pricing
	Sum of all ticket price	

5.Prompt the user to repeatedly to do the program(input (Yes or No)). If they response Yes go into the loop and prompt the user for county and market value of a home. Write a function to compute the assessed value. Pass to the function the county and market value. The function will determine the assessed value percent then compute and return the assessed value. (Multiple the market value by assessed value percent. Sum and display all market values and assessed values.

County/ Assessed Value Percent

Cook

0.90

DuPage

0.80

McHenry

0.75

Kane

0.60

All others

0.70

Input	Process	Output
Y/N	Pass county and market value to function	All market values
County Name	Assessed Value = Market Value * % (90/80/75/60/70) Cook/DuPage/McHenry/Kane /All others	All assessed values.
Market value of house	Sum all market values and assessed values.	

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